



MEDICAL APPOINTMENT NO SHOWS

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Introduction

This project analyzes the “Medical Appointment No Shows” dataset to understand patient attendance behavior and predict whether patients will attend or miss their appointments using machine learning and data visualization techniques.

Related Work

- Study 1: Found that ensemble methods (e.g., Random Forest) outperform base models, identifying “Waiting Time” as the most significant predictor of patient no-shows [1].
- Study 2: Demonstrated that integrating SMS notification data into predictive models (Decision Tree/Naive Bayes) significantly improves accuracy in forecasting attendance [2].

Objective

- Analyze medical appointment data.
- Predict patient no-show behavior.
- Identify factors affecting attendance.
- Improve healthcare scheduling and resource allocation.

Data

The dataset used in this project was obtained from Kaggle and contains medical appointment records. It includes patient information, appointment details, SMS reminders, and attendance status, helping analyze the main factors affecting no-show behavior in healthcare services.

Field	Description
Title	Medical Appointment No Shows
Storage	Kaggle Dataset
Owner	Joni Arroba
Target	To analyze patient appointment attendance patterns and identify the main factors affecting missed medical appointments and delays.
Meaning	[PatientId, AppointmentID, Gender, ScheduledDay, AppointmentDay, Age, Neighborhood, Scholarship, Hypertension, Diabetes, Alcoholism, Handicap, SMS_received, NoShow, WaitingDays]
Restriction	Accessible only to the project members during the project duration.
Date of Uploading Original Data	2016
Date Updated	Updated and processed by the team
Operation	Exploratory Data Analysis, Data Cleaning, Data Visualization, Feature Engineering
Number of Features	Before: 14 After Cleaning: 15
Number of Records	110,527 records

Table 1: Data Metadata and Description
Provides a detailed overview of the dataset attributes, including the source, the total number of records (110,527), and the features used for analysis after data cleaning.

Methodology

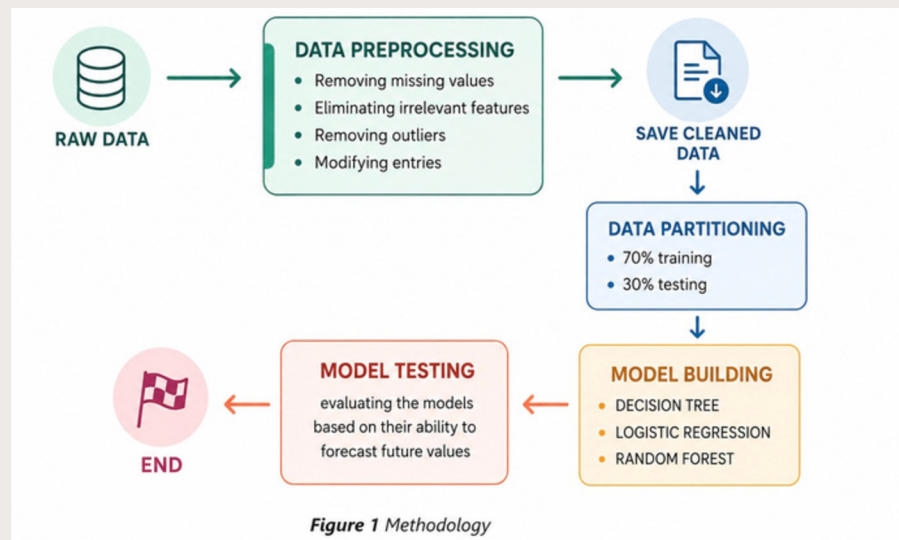


Figure 1: Methodology Workflow
Illustrates the project pipeline, including data preprocessing, data partitioning, and the model building and evaluation stages.

Goals

- Reduce patient no-show rates.
- Improve healthcare service efficiency.
- Discover patterns in patient behavior.
- Build predictive machine learning models.

Research Questions

- Can machine learning models predict patient no-show behavior accurately?
- Which factors affect appointment attendance the most?
- Does receiving SMS reminders improve attendance?
- Do longer waiting days increase missed appointments?

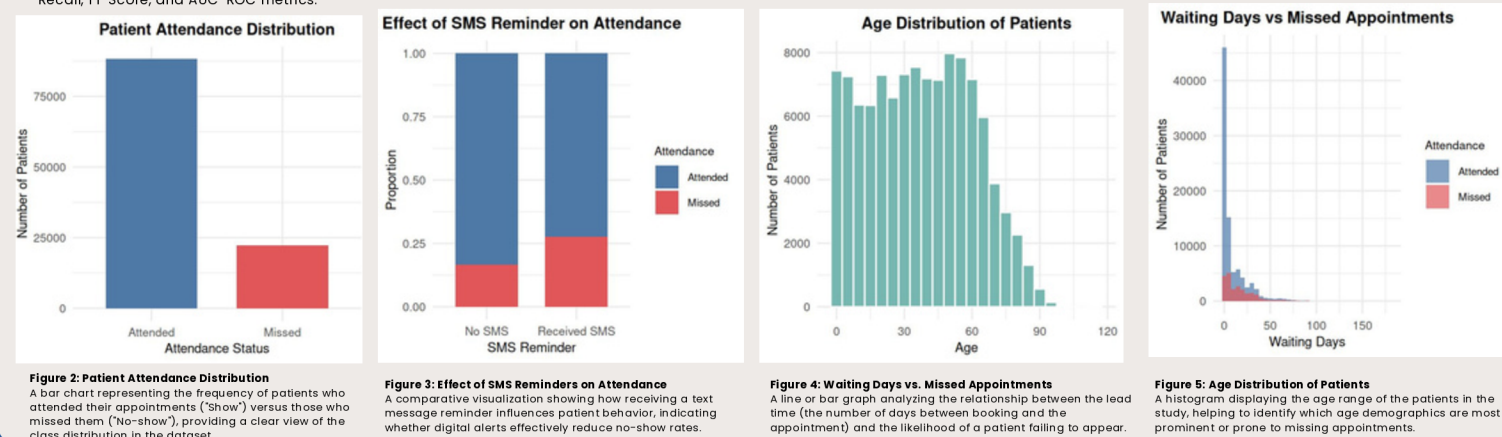
Hypotheses

- H1:
Machine learning models can effectively predict patient no show behavior using demographic and medical related features.
- H2:
Ensemble learning methods such as Random Forest provide stable and reliable prediction performance for no show classification.

Results

MODEL ALGORITHM	ACCURACY	PRECISION	RECALL	F1-SCORE	AUC	ROC
Decision Tree	80.3%	80.3%	100%	0.8907	0.793	0.790
Random Forest	79.8%	80.4%	98.9%	0.8872	0.852	0.850
Logistic Regression	80.0%	80.3%	99.5%	0.8889	0.756	0.760
Naive Bayes	64.4%	56.6%	87.0%	0.6860	0.727	0.730

Table 2: Performance Evaluation of Models
Summarizes the comparative results of the machine learning algorithms (Decision Tree, Random Forest, Logistic Regression, and Naive Bayes) based on Accuracy, Precision, Recall, F1-Score, and AUC-ROC metrics.



Conclusion

The results demonstrate that machine learning models can help predict whether patients will attend or miss their appointments, which may contribute to improving appointment scheduling efficiency, reducing no-show rates, and supporting decision-making in the healthcare sector.

Future Work

- Use larger and more recent healthcare datasets.
- Apply advanced deep learning models.
- Include additional patient behavior variables.
- Develop real-time prediction systems for hospitals.
- Improve model accuracy using feature engineering and optimization techniques.

References

- [1] Kaggle, "Medical Appointment No Shows Dataset," 2016. [Online]. Available: <https://www.kaggle.com/datasets/joniarroba/noshowappointments>
- [2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
- [3] W. McKinney, "Data Structures for Statistical Computing in Python," in Proceedings of the 9th Python in Science Conference, 2010, pp. 51–56.
- [4] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.
- [5] M. A. Haq and M. A. R. Khan, "Machine Learning Approaches to Predict No-Shows in Saudi Rehabilitation Hospitals," Saudi J. Health Syst. Res., vol. 5, no. 1, pp. 24–32, 2025.
- [6] G. Aljuhani, "Predicting Medical Appointment No-Shows using Data Mining Techniques," in Data Visions Virtual Expo, 2025.